Research Introduction
Transforming 8x8 Pixel Art to 40x40 Using Diffusion Models

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Introduction

In recent years, image generation using Diffusion Models has attracted significant attention.

These models typically use text or images as input.

Using text as input for image generation:
there can be discrepancies between the generated image and the intended image.

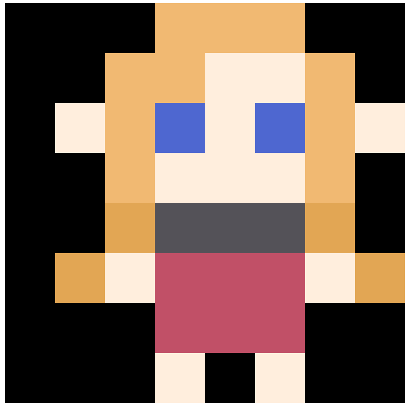
Using images as input:
It is possible to generate images that closely match the desired result.
However, it is difficult to prepare appropriate input images.

Introduction

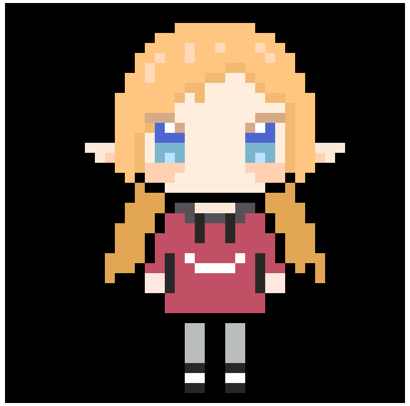
In this study, we propose a method for image generation using 8x8 pixel art, which can be easily created by anyone, as input.

Specifically, we aim to generate 40x40 pixel art.

input

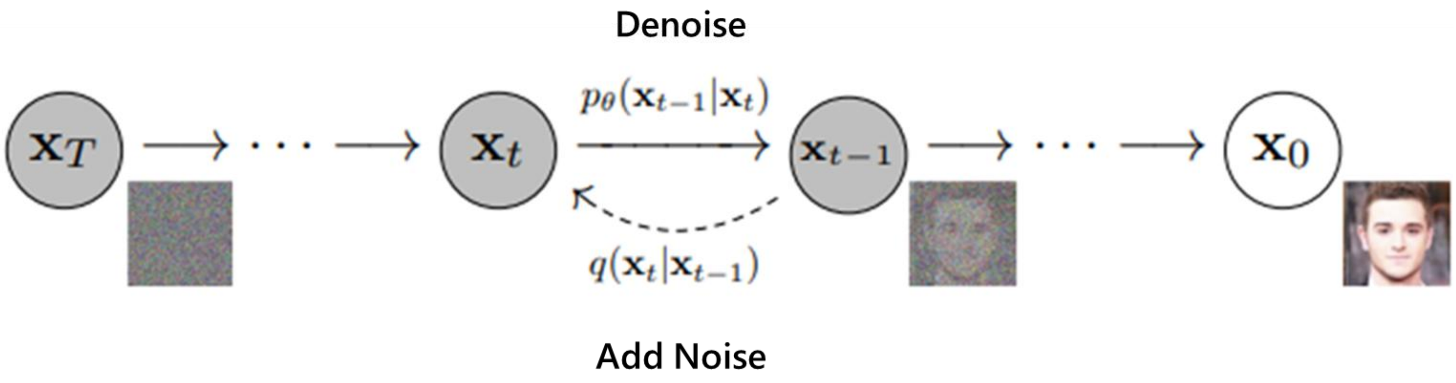


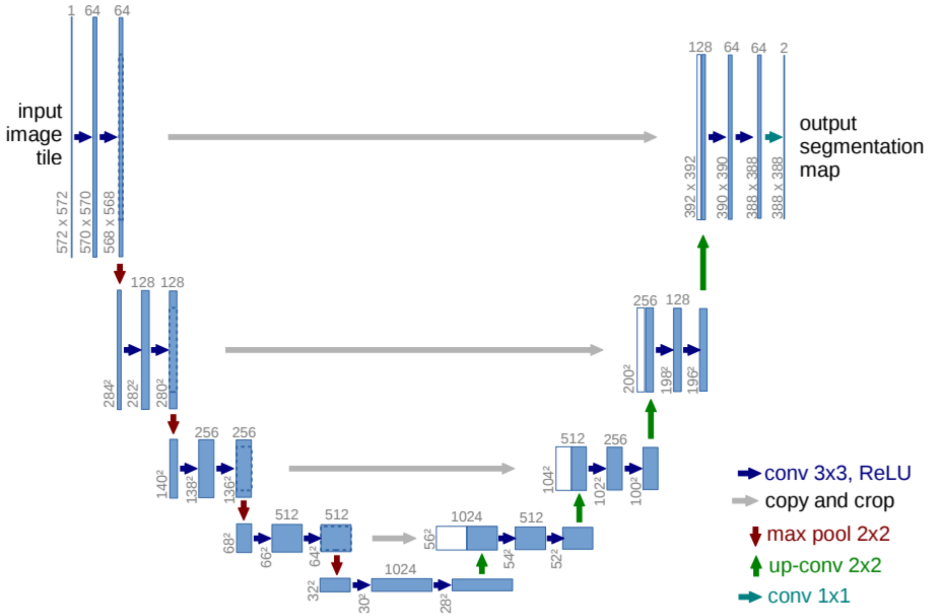
target



Method

DiffusionModel(DDPM)





Take a sample x_0 from the data, choose an arbitrary timestep t, and generate Gaussian noise ϵ.
Using x_0 and ϵ, create noisy data $\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$.
Input this to the network ϵ_θ, and calculate the error between the predicted noise and the actual noise.
Update the network parameters using gradient descent to minimize this error.

Algorithm 1 Training

repeat
  $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
  $t \sim \mathrm{Uniform}(\{1, \dots, T\})$
  $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Take gradient descent step on
  $$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
until converged

Dataset

Prepared a training dataset of 50 images.

To expand the dataset, I performed ColorTransfer.
To prevent the influence of the background, I used only the ab components in the Lab color space.
The method used is the common approach of matching the input colors to the target by using the standard deviation of the ab components.

The number of images after expansion is 2500.

input target output

Dataset URL

https://drive.google.com/drive/folders/1SwHgpKc
FisSWCM86BH8WNrzXkEgH4_40?usp=sharing

Preliminary Experiment

Since we are generating relatively small images, specifically 40×40 pixels,
it may not be necessary to use as many channels as those employed in existing models.

Compare the models

Pix2Pix

Diffusion Model

Diffusion Model with reduced channels

Preliminary Experiment

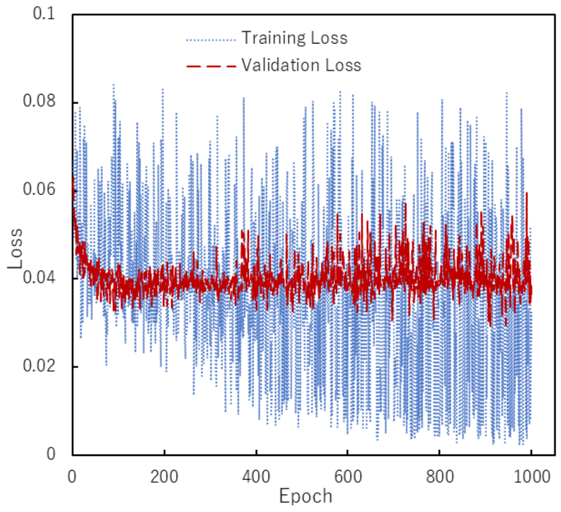
Train Diffusion models and Pix2Pix

Train Condition

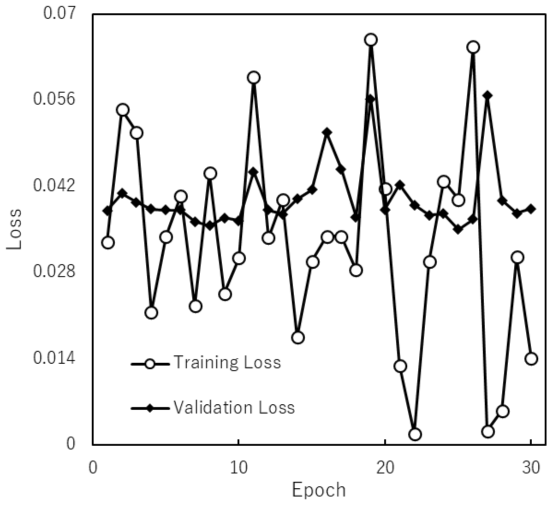
No.	1	2	3	4
Model	Diffusion	Diffusion	Diffusion	Pix2Pix
Train	50set	2500set	2500set	2500set
Validation	1set	1set	1set	1set
Unet Channel	64,128,256,512	64,128,256,512	64,64,64,64	64,128,256
Batch size	8	8	8	8
Epoch	1000	1000	1000	1000

Result

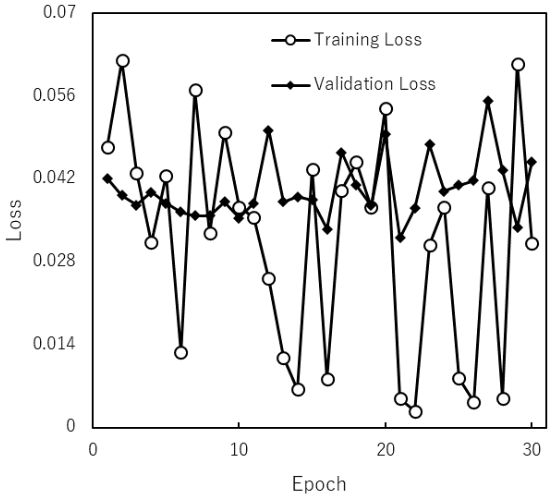
1



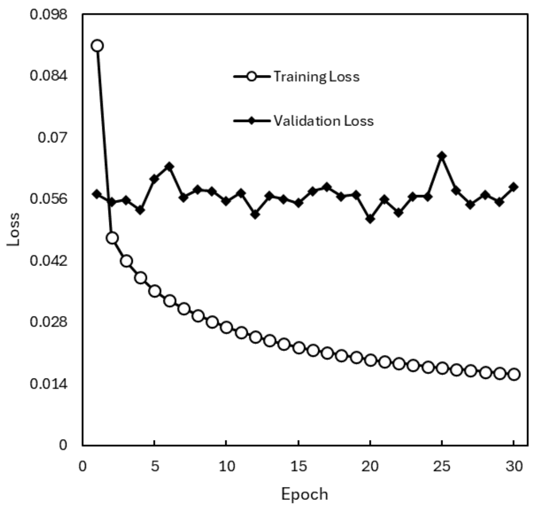
2



3



4



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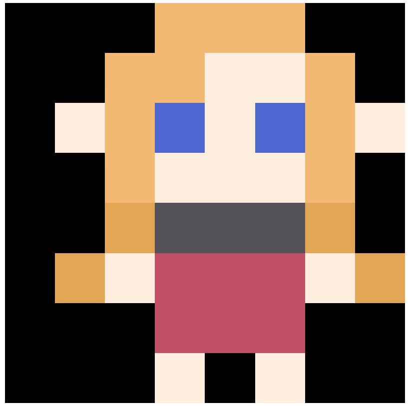
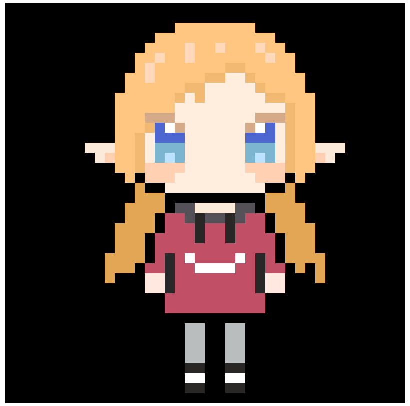
Result

input



target



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Output

Diffusion

1

2

3

pix2pix

4

Future Work

Dataset Expansion

- increase the number of pixel art of people.
- expand to other types of pixel art.

Model Parameter Optimization

- Vision Transformer for Denoise Network